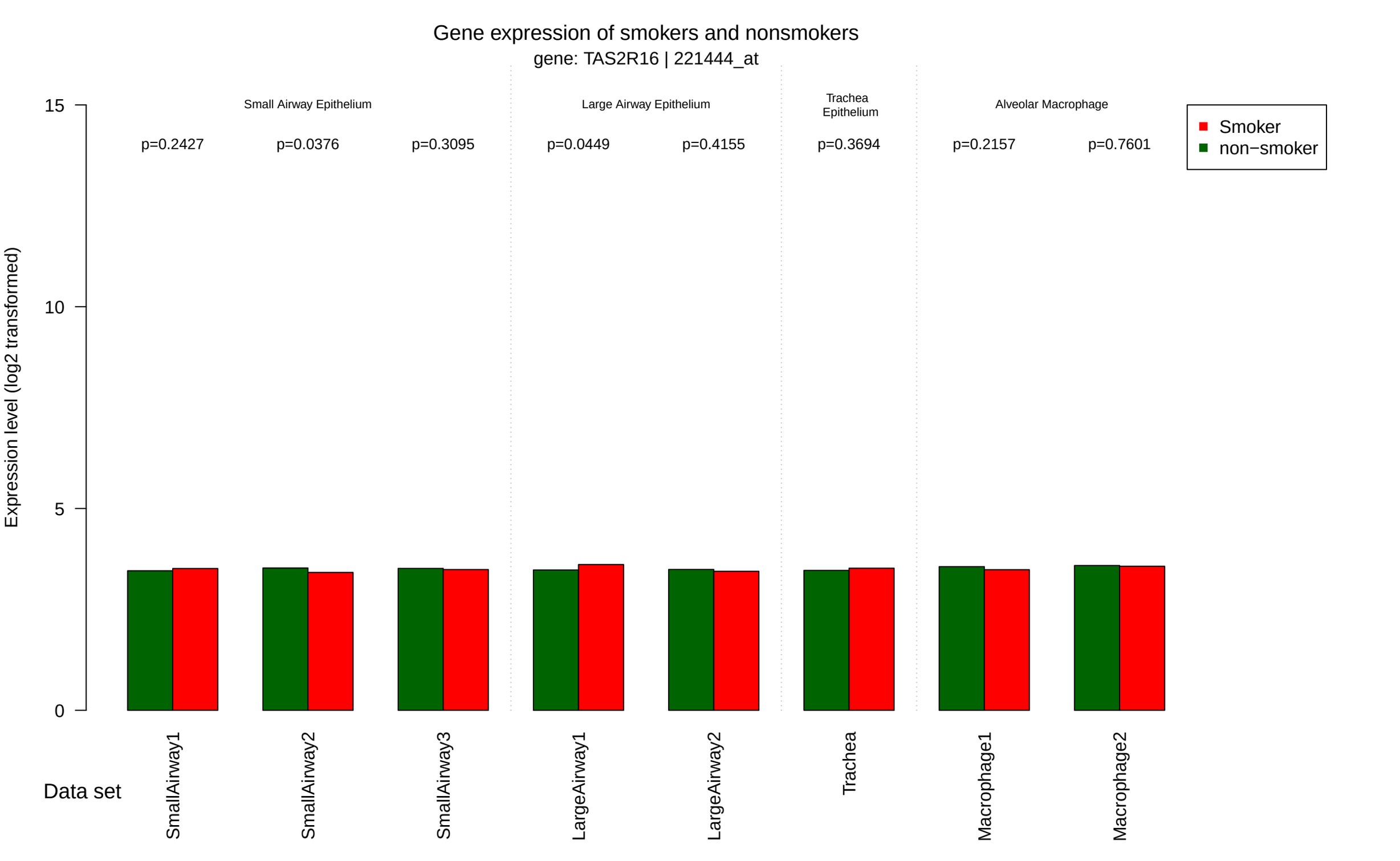
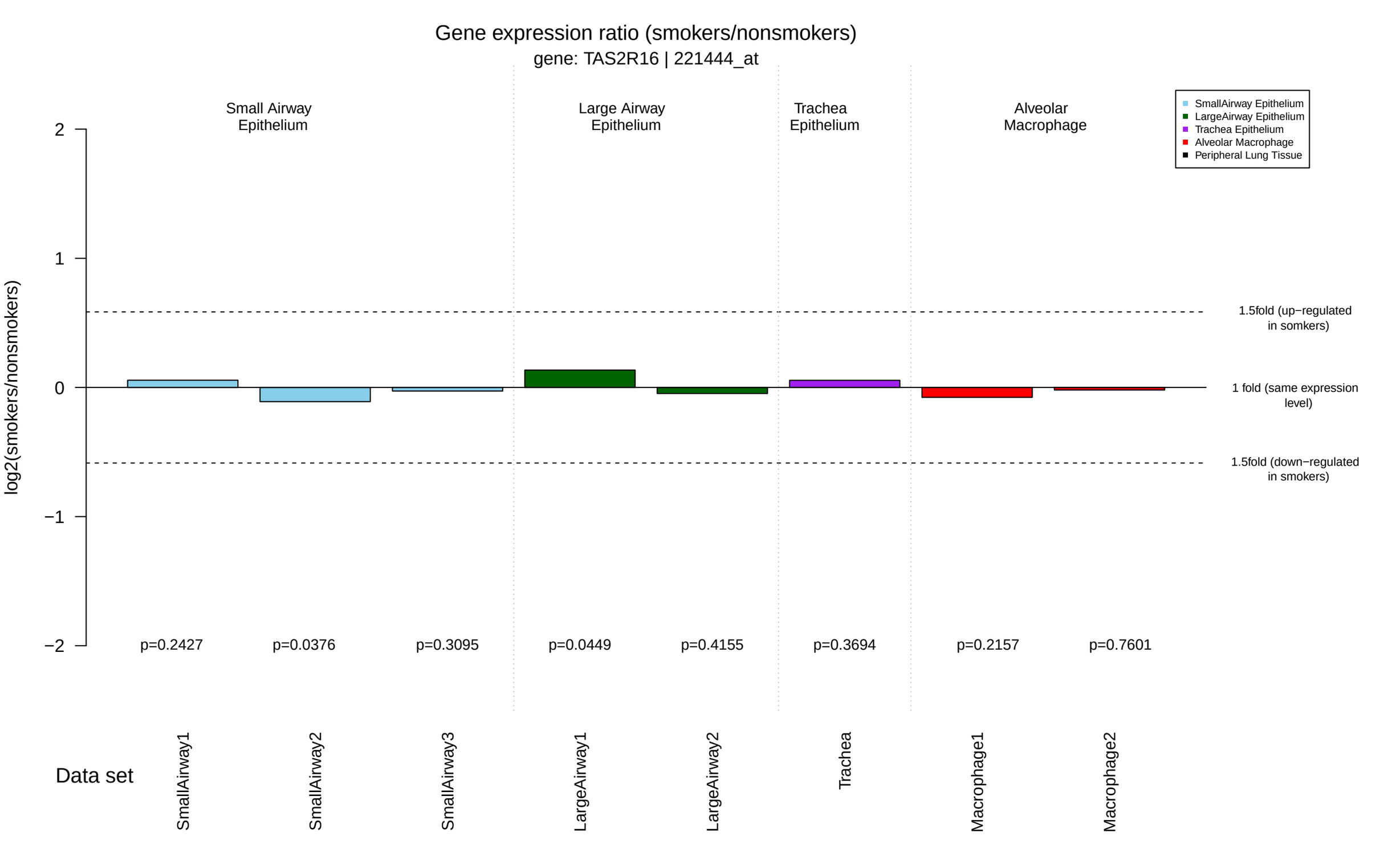
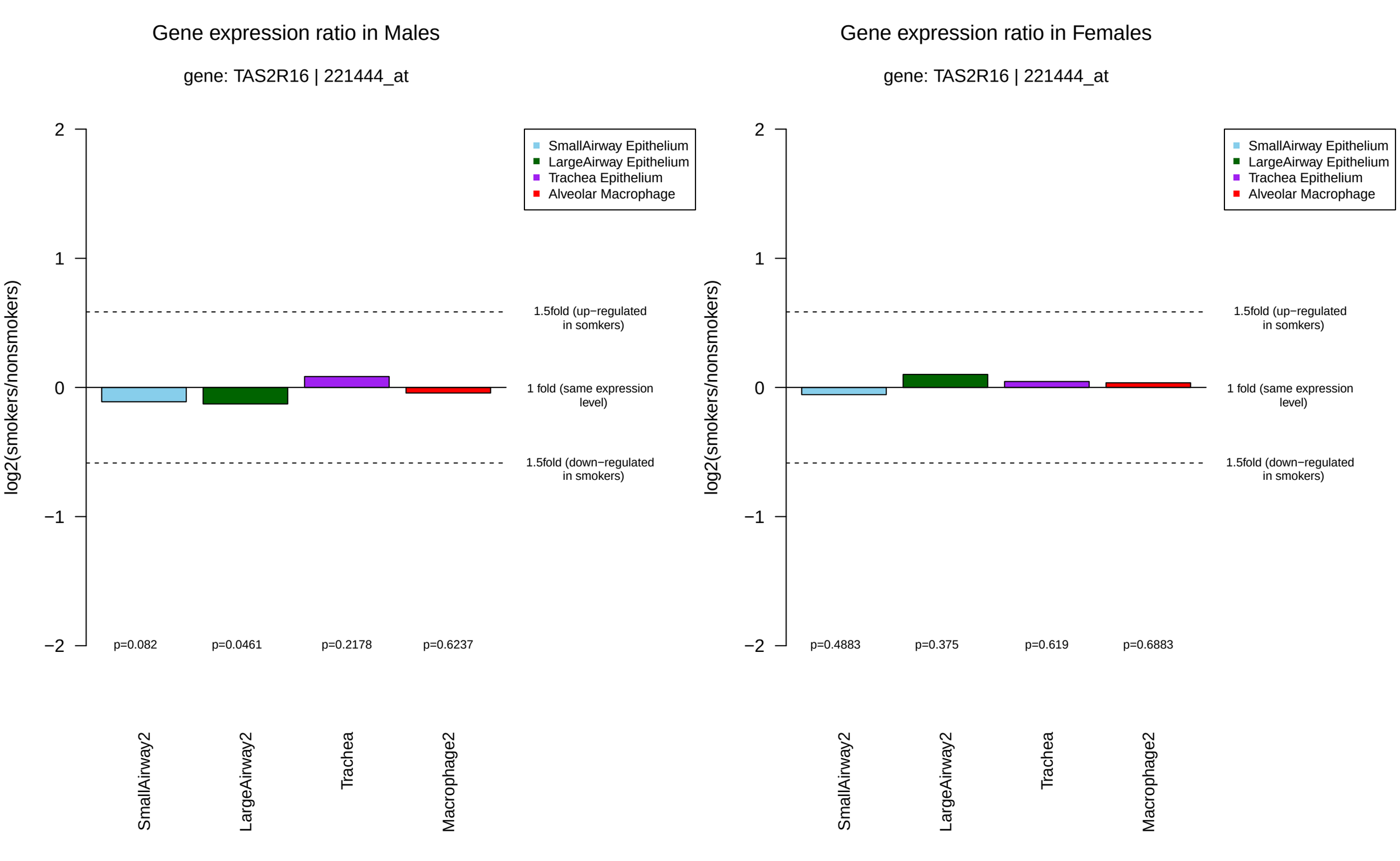
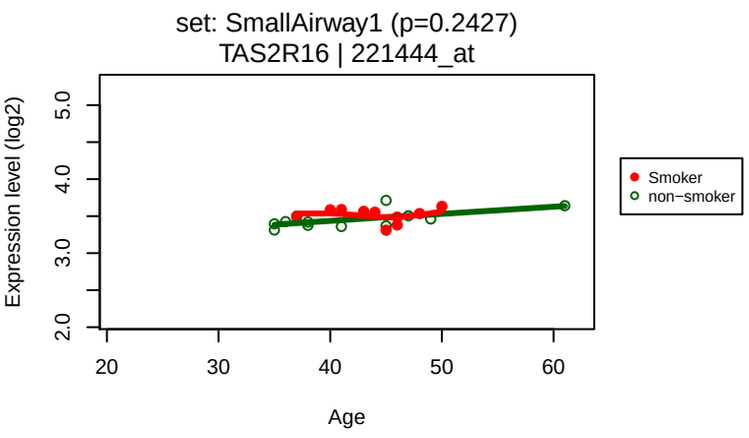
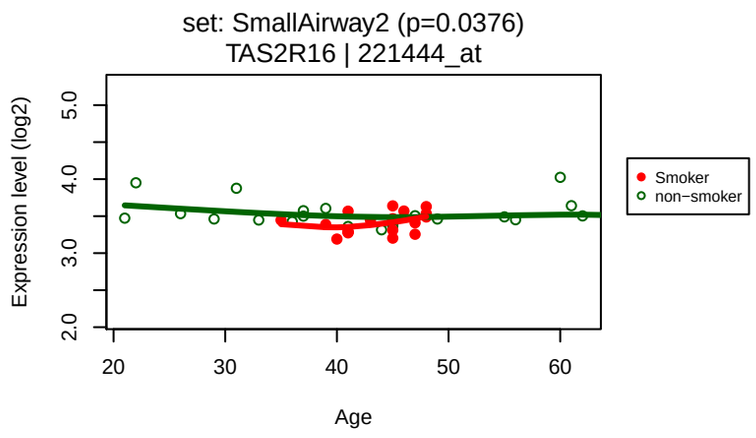
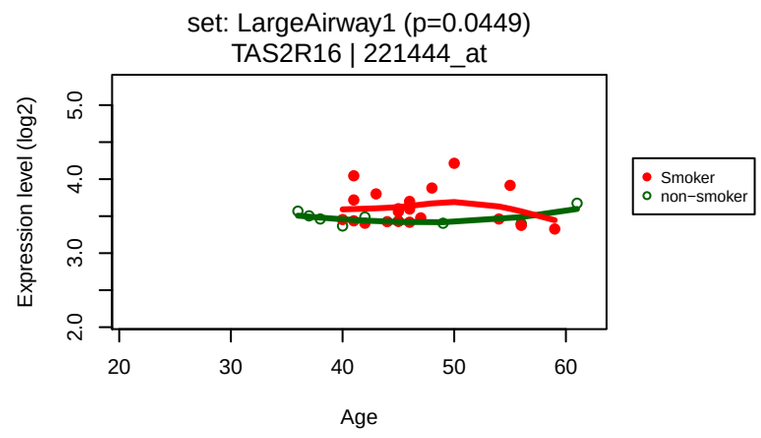

Gene expression of smokers and nonsmokers
gene: TAS2R16 | 221444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2427
p=0.0376
p=0.3095
p=0.0449
p=0.4155
p=0.3694
p=0.2157
p=0.7601
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

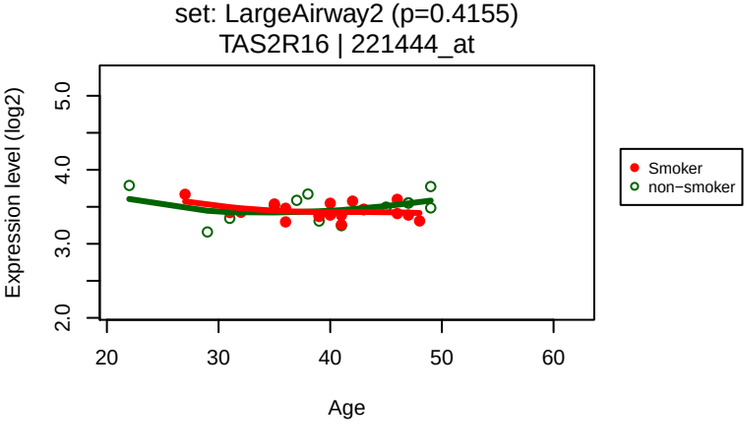

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R16 | 221444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2427
p=0.0376
p=0.3095
p=0.0449
p=0.4155
p=0.3694
p=0.2157
p=0.7601
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

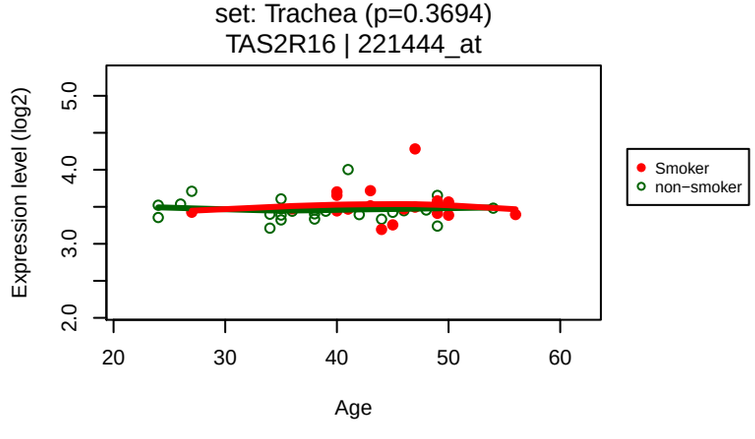

Gene expression ratio in Males
gene: TAS2R16 | 221444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.082
p=0.0461
p=0.2178
p=0.6237
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R16 | 221444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4883
p=0.375
p=0.619
p=0.6883
SmallAirway2
LargeAirway2
Trachea
Macrophage2

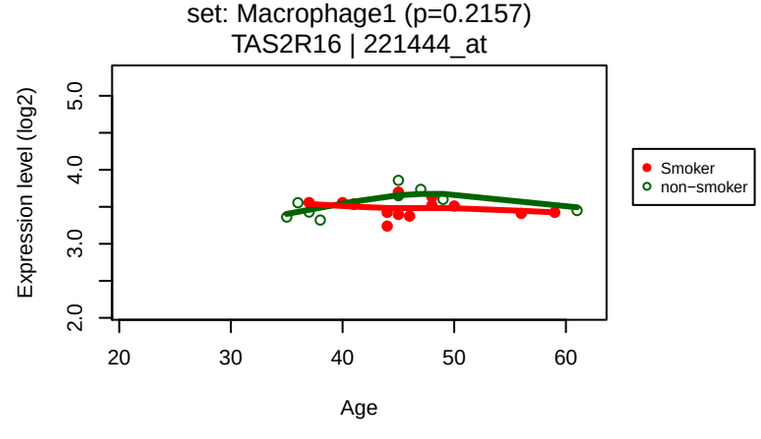

set: SmallAirway1 (p=0.2427)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker

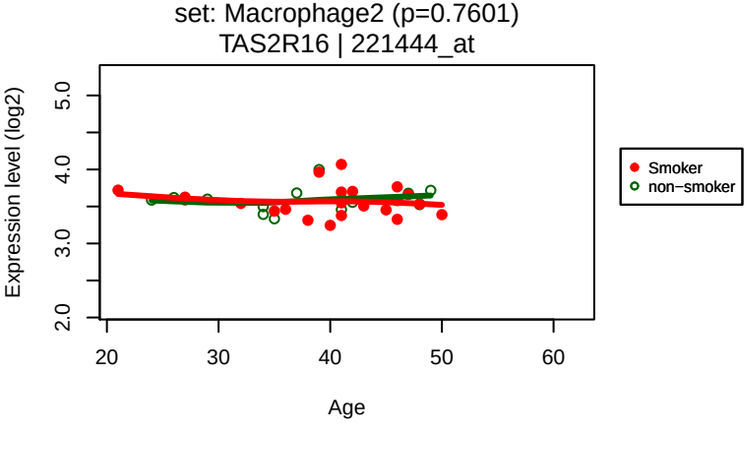

set: SmallAirway2 (p=0.0376)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker

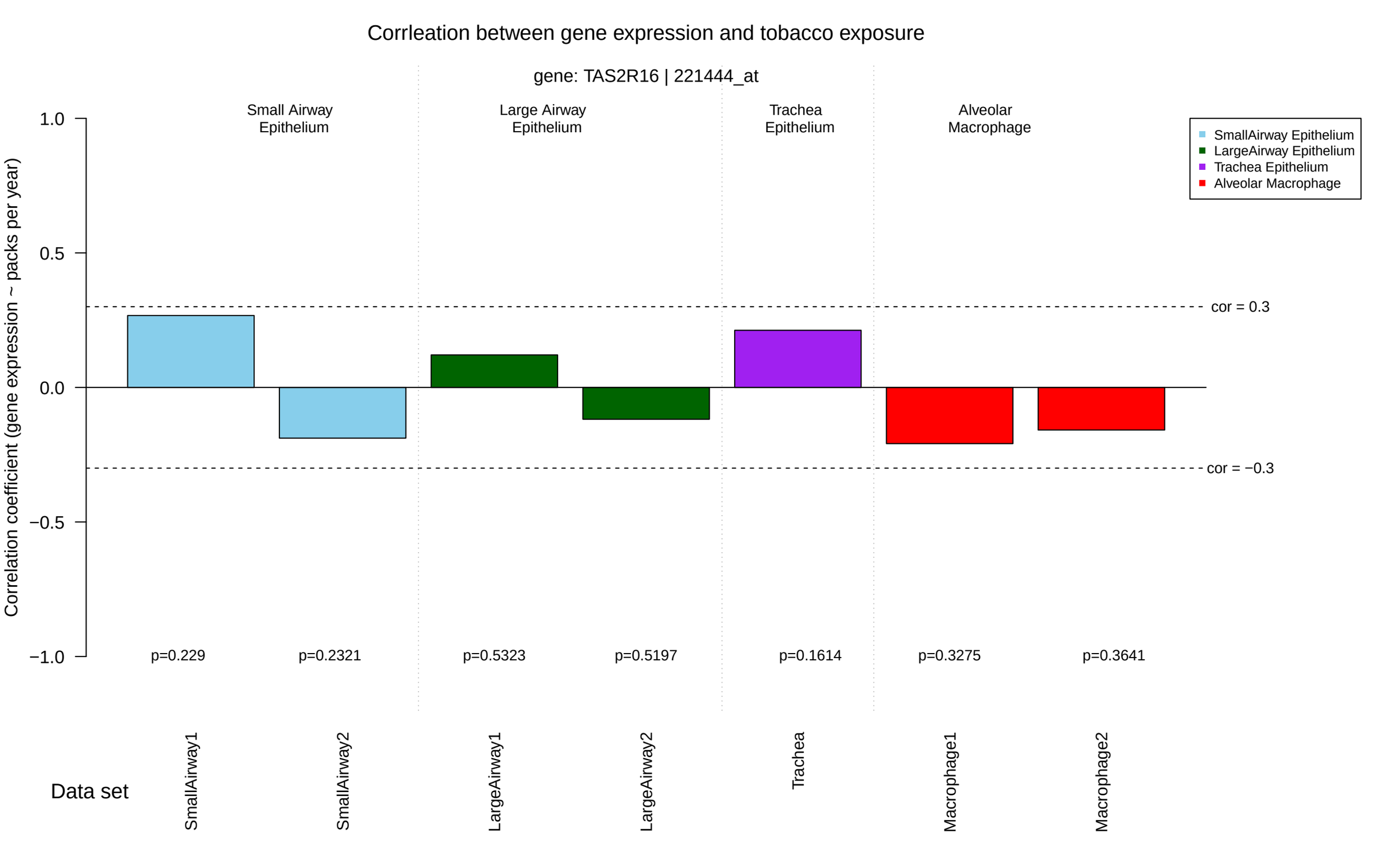

set: LargeAirway1 (p=0.0449)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4155)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3694)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2157)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7601)
TAS2R16 | 221444_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R16 | 221444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.229
p=0.2321
p=0.5323
p=0.5197
p=0.1614
p=0.3275
p=0.3641
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2